CHAPTER – 3

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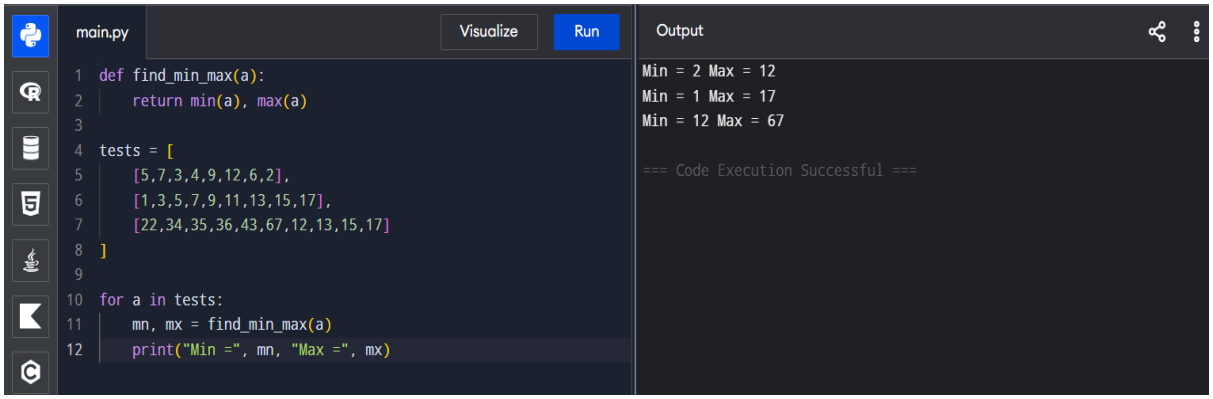
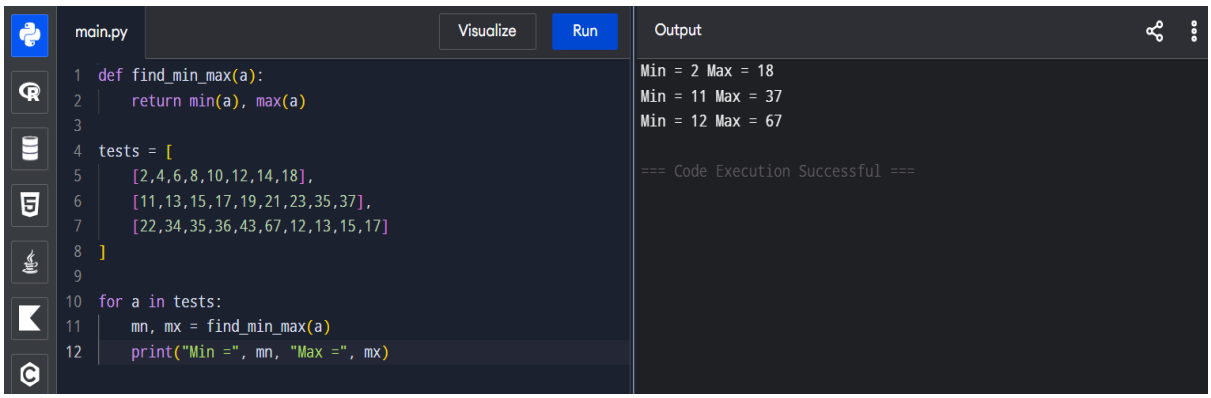
Course: CSA0603– DAA

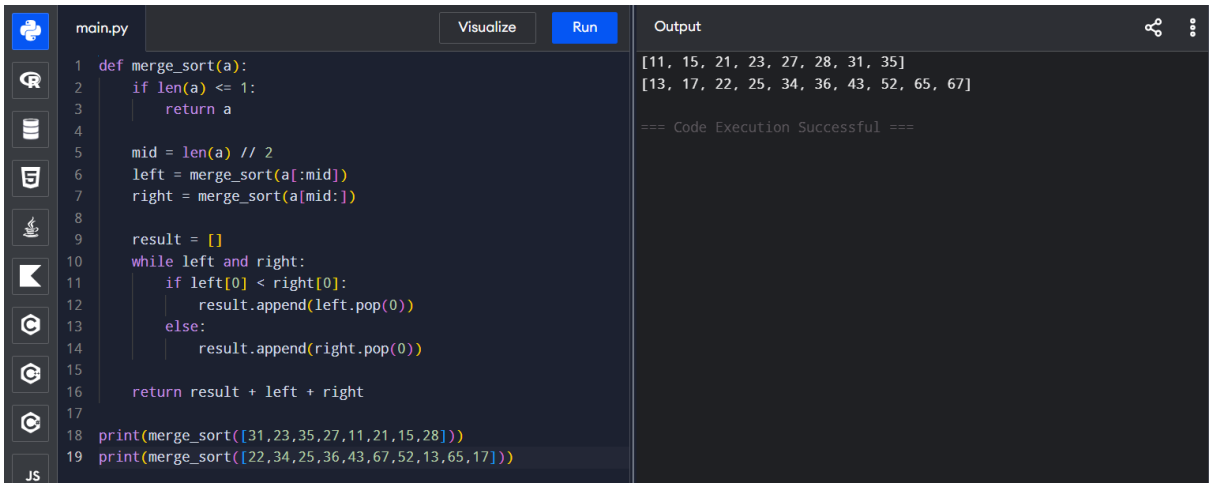
EXPERIMENT – 1



EXPERIMENT – 2



EXPERIMENT – 3



EXPERIMENT – 4

main.py

```python
def merge_sort(a):
    global count
    if len(a) <= 1:
        return a
    m = len(a)//2
    left = merge_sort(a[:m])
    right = merge_sort(a[m:])
    result = []
    while left and right:
        count += 1
        if left[0] <= right[0]:
            result.append(left.pop(0))
        else:
            result.append(right.pop(0))
    return result + left + right
tests = [
    [12,4,78,23,45,67,89,1],
    [38,27,43,3,9,82,10]
]
for a in tests:
    count = 0
    print("Sorted:", merge_sort(a))
```

Output

```
Sorted: [1, 4, 12, 23, 45, 67, 78, 89]
Comparisons: 16
Sorted: [3, 9, 10, 27, 38, 43, 82]
Comparisons: 13

=== Code Execution Successful ===
```

EXPERIMENT – 5

main.py

```python
def quick_sort(a):
    if len(a) <= 1:
        return a
    pivot = a[0]
    left = [x for x in a[1:] if x <= pivot]
    right = [x for x in a[1:] if x > pivot]
    print("Pivot:", pivot, "->", left + [pivot] + right)
    return quick_sort(left) + [pivot] + quick_sort(right)
tests = [
    [10,16,8,12,15,6,3,9,5],
    [12,4,78,23,45,67,89,1],
    [38,27,43,3,9,82,10]
]
for a in tests:
    print("Sorted:", quick_sort(a))
    print()
```

Output

```
Pivot: 10 -> [8, 6, 3, 9, 5, 10, 16, 12, 15]
Pivot: 8 -> [6, 3, 5, 8, 9]
Pivot: 6 -> [3, 5, 6]
Pivot: 3 -> [3, 5]
Pivot: 16 -> [12, 15, 16]
Pivot: 12 -> [12, 15]
Sorted: [3, 5, 6, 8, 9, 10, 12, 15, 16]

Pivot: 12 -> [4, 1, 12, 78, 23, 45, 67, 89]
Pivot: 4 -> [1, 4]
Pivot: 78 -> [23, 45, 67, 78, 89]
Pivot: 23 -> [23, 45, 67]
Pivot: 45 -> [45, 67]
Sorted: [1, 4, 12, 23, 45, 67, 78, 89]

Pivot: 38 -> [27, 3, 9, 10, 38, 43, 82]
Pivot: 27 -> [3, 9, 10, 27]
Pivot: 3 -> [3, 9, 10]
Pivot: 9 -> [9, 10]
Pivot: 43 -> [43, 82]
Sorted: [3, 9, 10, 27, 38, 43, 82]
```

EXPERIMENT – 6

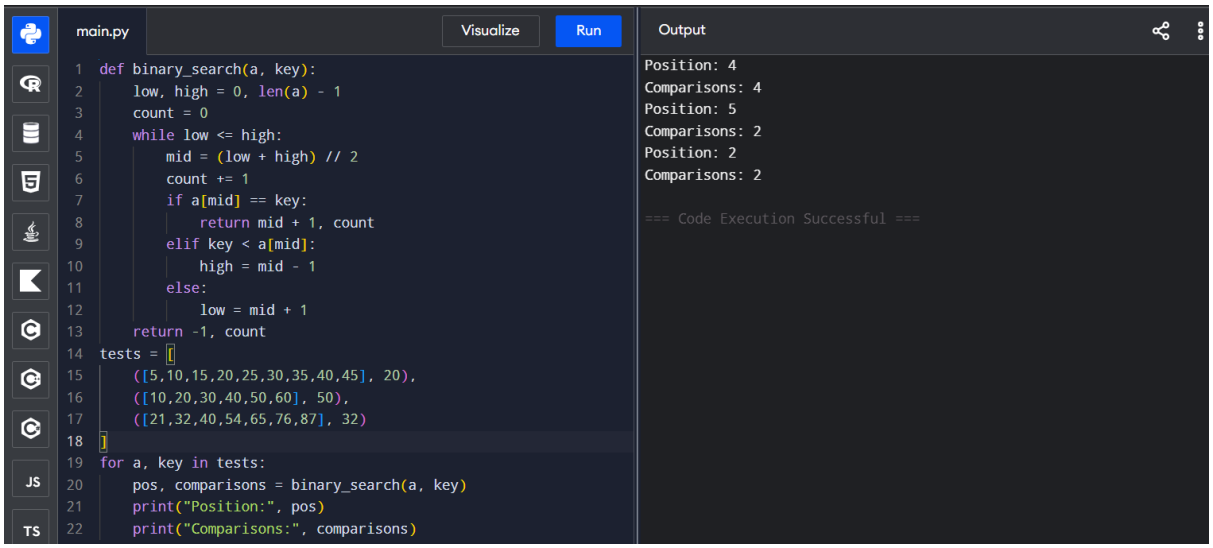
main.py

```python
def quick_sort(a):
    if len(a) <= 1:
        return a
    pivot = a[len(a)//2]
    left = [x for x in a if x < pivot]
    mid = [x for x in a if x == pivot]
    right = [x for x in a if x > pivot]
    print("Pivot:", pivot, "->", left + mid + right)
    return quick_sort(left) + mid + quick_sort(right)
tests = [
    [19,72,35,46,58,91,22,31],
    [31,23,35,27,11,21,15,28],
    [22,34,25,36,43,67,52,13,65,17]
]
for a in tests:
    print("Sorted:", quick_sort(a))
    print()
```

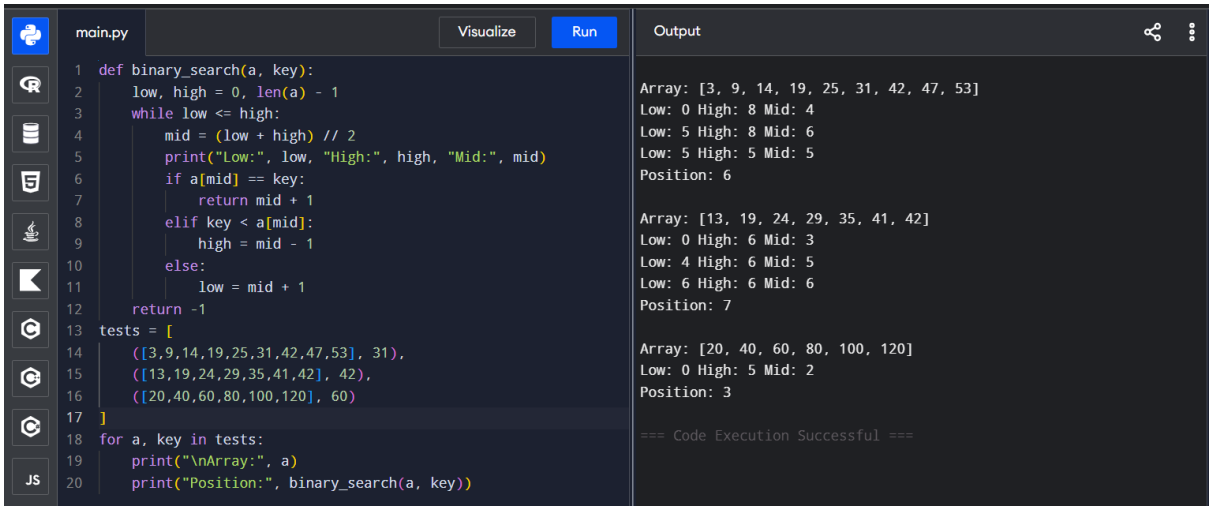
Output

```
Pivot: 58 -> [19, 35, 46, 22, 31, 58, 72, 91]
Pivot: 46 -> [19, 35, 22, 31, 46]
Pivot: 22 -> [19, 22, 35, 31]
Pivot: 31 -> [31, 35]
Pivot: 91 -> [72, 91]
Sorted: [19, 22, 31, 35, 46, 58, 72, 91]

Pivot: 11 -> [11, 31, 23, 35, 27, 21, 15, 28]
Pivot: 27 -> [23, 21, 15, 27, 31, 35, 28]
Pivot: 21 -> [15, 21, 23]
Pivot: 35 -> [31, 28, 35]
Pivot: 28 -> [28, 31]
Sorted: [11, 15, 21, 23, 27, 28, 31, 35]

Pivot: 67 -> [22, 34, 25, 36, 43, 52, 13, 65, 17, 67]
Pivot: 43 -> [22, 34, 25, 36, 13, 17, 43, 52, 65]
Pivot: 36 -> [22, 34, 25, 13, 17, 36]
Pivot: 25 -> [22, 13, 17, 25, 34]
Pivot: 13 -> [13, 22, 17]
Pivot: 17 -> [17, 22]
Pivot: 65 -> [52, 65]
Sorted: [13, 17, 22, 25, 34, 36, 43, 52, 65, 67]
```

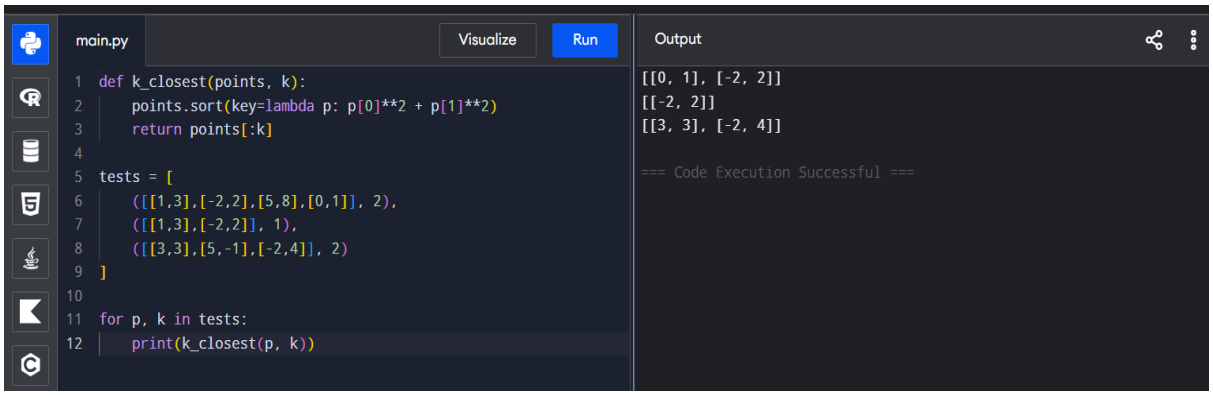
EXPERIMENT – 7



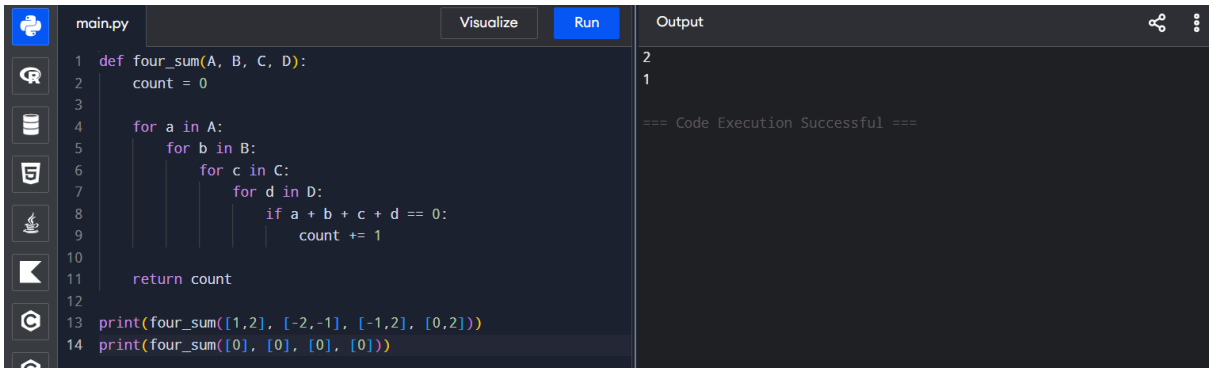
EXPERIMENT – 8



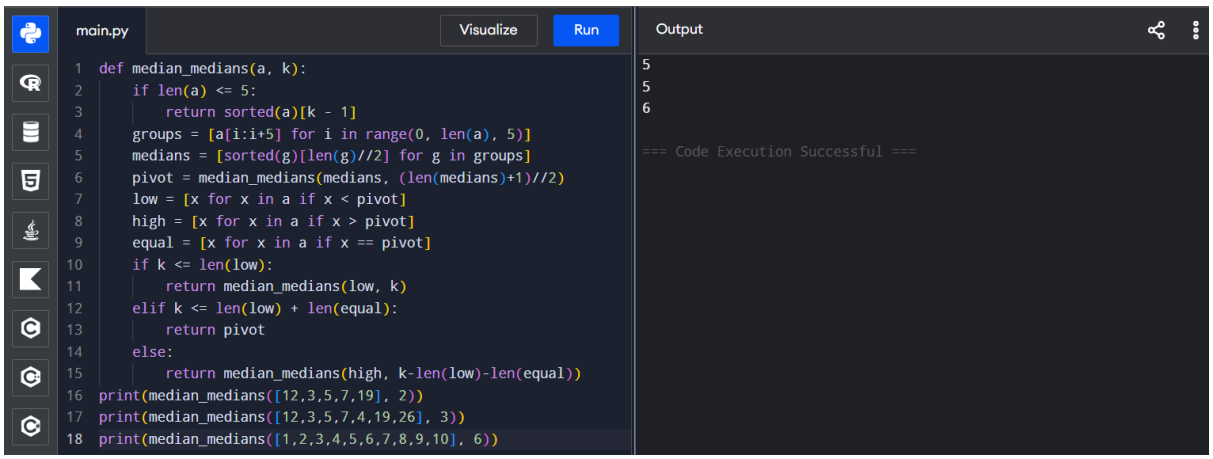
EXPERIMENT – 9



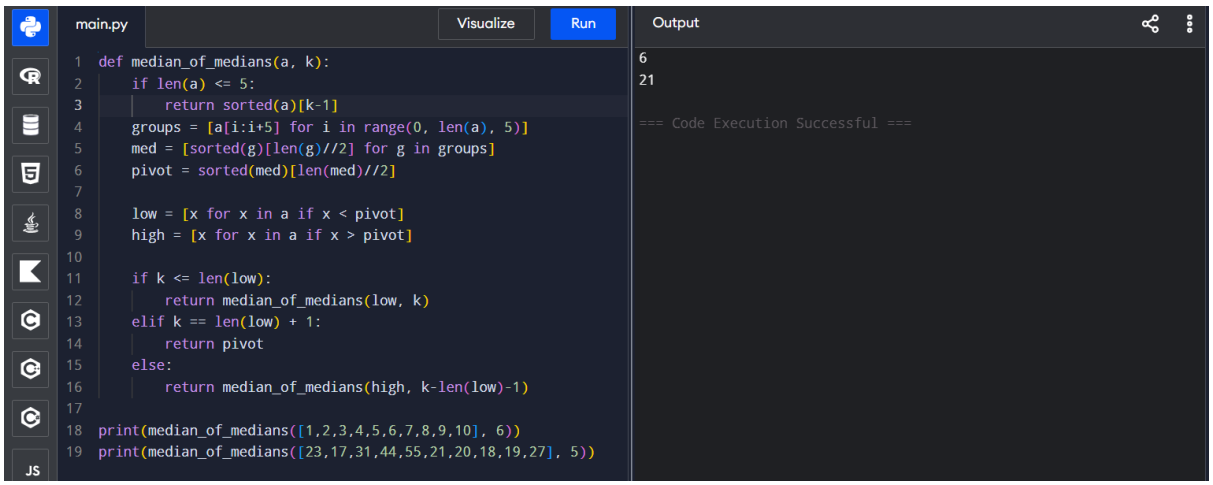
EXPERIMENT – 10



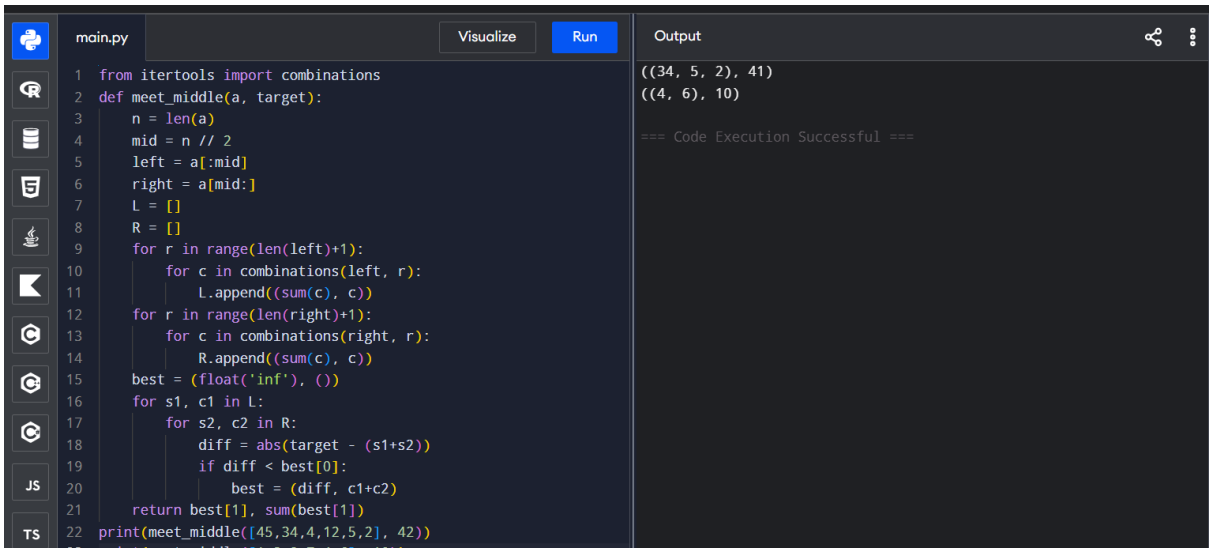
EXPERIMENT – 11



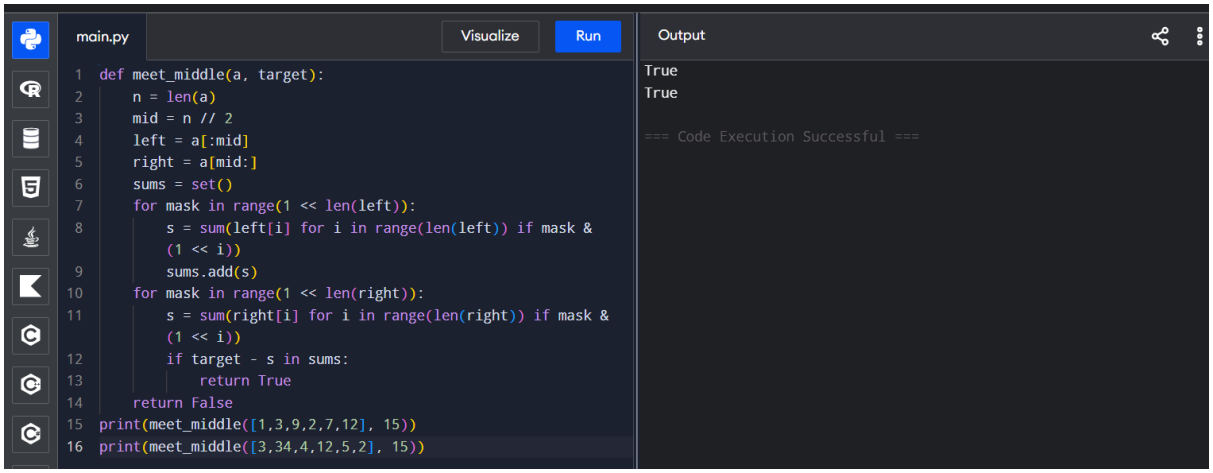
EXPERIMENT – 12



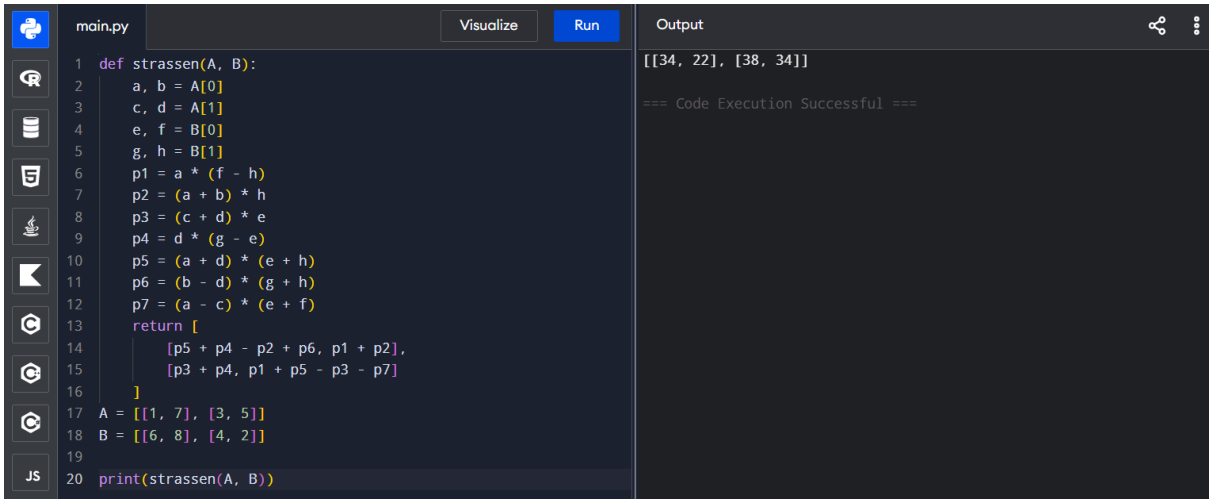
EXPERIMENT – 13



EXPERIMENT – 14



EXPERIMENT – 15



EXPERIMENT – 16